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# Complex spatial and temporal variation of subtropical benthic macrofauna under sewage impact

F.M. Souza [a, *], K.M. Brauko [a, b], E.R. Gilbert [a], C.C. Martins [a], P.C. Lana [a], M.G. Camargo [a, c]

[a] Centro de Estudos do Mar, Universidade Federal do Paraná, 83255-976 Pontal do Paraná, Paraná, Brazil
[b] Núcleo de Estudos do Mar, CCB, Universidade Federal de Santa Catarina, 88040-900 Florianópolis, Santa Catarina, Brazil
[c] Instituto de Oceanografia, Universidade Federal do Rio Grande, 96205080 Rio Grande, Rio Grande do Sul, Brazil

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ABSTRACT

To better understand variation in sewage-impacted benthic macrofauna from subtropical tidal flats over time and space, we applied a five-factor linear model at a hierarchy of spatial (Condition – Contaminated or Non-Contaminated, Tidal Flat and Plot) and temporal scales (Season and Fortnight). The Contaminated site showed high levels of coprostanol and the presence of *Paranais* cf *frici* as markers or indicators of organic enrichment. Differences between Seasons were more pronounced for the faunal variation patterns than for the other parameters, with lower species richness and abundance in summer. There were significant interactions between Fortnight and Tidal Flat for most variables, reflecting marked heterogeneity within Tidal Flats. Benthic community has significantly changed over short periods of time. These rapid changes may lead to erroneous interpretations and mask the true sources of variation. Our results clearly demonstrate the need to better understand benthic temporal variability even at small scales.
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## 1. Introduction

Benthic infauna are widely used as indicators of water and sediment quality due to the variable degrees of sensitivity and tolerance of individual species or assemblages (Borja and Dauer, 2008; Dauer et al., 2000; Lancellotti and Stotz, 2004). Macrobenthic assemblages are known to vary at many spatial and temporal scales, which may range from days to years and from centimeters to dozens of kilometers (Bessa et al., 2014; Díaz-Tapia et al., 2013; França et al., 2009). To better understand macrofaunal distribution patterns and their driving factors, experimental studies must incorporate adequate replication at multiple nested scales (Chapman et al., 2009). Experimental designs without adequate replication may mask the true sources of variation and may lead to erroneous interpretations. Replicated sampling across different scales is an efficient tool for understanding structural variation in ecological communities and the background sources of variation, which are often confounded with community responses to anthropogenic disturbance (de Knegt et al., 2010; Underwood, 1997).

Hierarchical models are increasingly used in marine environments to assess the scales at which species and communities vary (Muniz et al., 2012; Williams et al., 2015). Nevertheless, relationships between temporal and spatial variability patterns of living and non-living components of ecological systems are still not well understood. Improving the understanding of such variability would allow to distinguish between changes caused by human activities and the natural variability, leading to a more realistic evaluation of environmental impacts (Souza et al., 2013; Thrush et al., 1999; Underwood and Chapman, 2013).

Environmental pollution creates extensive and urgent problems for the conservation and restoration of degraded habitats. Solving these problems requires the application of high-quality scientific research, including the use of improved sampling designs for assessing faunal patterns of variation associated with geochemical markers of the impacts themselves (Underwood and Chapman, 2013).

Fecal sterols (coprostanol and epicoprostanol) are widely used as markers to assess levels of sewage contamination. Coprostanol (5β(H)-cholestan-3β-ol) is produced during the degradation of cholesterol in digestive tracts of higher vertebrates, including humans. It remains in the feces and, consequently, in sewage inputs (Grimalt et al., 1990; Readman et al., 2005; Martins et al., 2008). Epicoprostanol (5β(H)-cholestan-3α-ol) is formed during the anaerobic treatment of wastewater and can be used to assess the

* Corresponding author.
*E-mail address:* fernanda.cem@gmail.com (F.M. Souza).

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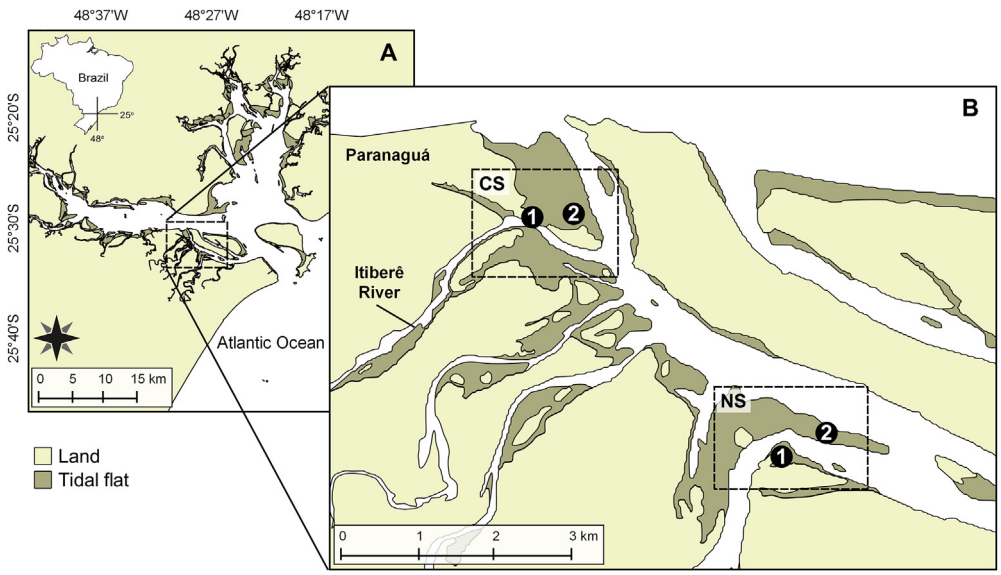


**Fig. 1.** Paranaguá Bay (A) and Cotinga sub-estuary (B). Tidal flats 1 and 2, sampled at the Contaminated (CS) and Non-Contaminated sites (NS).

level of treatment applied to the fecal matter in sewage before discharge (Mudge and Seguel, 1999). These markers are used as alternatives to microorganism analyses such as fecal coliform analysis because they are organic compounds that accumulate in marine sediments and are less susceptible to environmental degradation than other sewage contamination indicators (Carreira et al., 2004; Martins et al., 2014).

In this context, spatio-temporal variability patterns of benthic assemblages should fluctuate according to the scales at which the sewage discharges affects a given system, and this variation will differ between Contaminated and Non-Contaminated habitats. To test this assumption, we have assessed the structural responses of benthic fauna to key contamination drivers at tidal flats with varying levels of sewage markers along a subtropical estuary from southern Brazil, by applying a sampling design hierarchized in time and space.

## 2. Materials and methods

### 2.1. Study area

The Paranaguá Bay, centered 25°03′S, 48°25′W, is one of the largest and best preserved estuarine systems in southern South America (Lana et al., 2001). Despite the bay keeps some pristine sectors, largely protected by conservation units, human impacts have grown over the last decades, mainly associated with port activities, agricultural activities and the input of domestic and industrial sewage (Lana et al., 2001; Martins et al., 2010).

Sampling was carried out at the Cotinga polyhaline sub-estuary, a channel surrounded by mangroves forests and salt marshes, close to Paranaguá city (Fig. 1). The sub-estuary has a surface area of 21.38 km$^2$, of which 34% made up by tidal flats (Noernberg et al., 2006). One of the main tributaries of the Cotinga sub-estuary is the Itiberê River, into which urban sewage and effluents from the city of Paranaguá are discharged.

According to the local water and sanitation company, only 50% of the Paranaguá's wastewater is treated, and the remaining is discharged *in natura*. A clear gradient of organic pollution is evidenced by high concentrations of fecal sterols (Abreu-Mota et al., 2014; Martins et al., 2010), high bacterial activity (Kolm et al., 2002) and abundance of macrobenthic bioindicators of organic enrichment (Brauko et al., 2015; Souza et al., 2013) near the Itiberê River.

### 2.2. Sampling design and sample processing

The sampling design was based on a five-factor model (two temporal and three spatial factors). Temporal factors included Season (Summer and Winter) and three consecutive Fortnights within each Season. Spatial factors included Condition (Contaminated and Non-Contaminated – $10^3$; m apart), with each Condition, sampled in two Tidal Flats ($10^2$ m apart) at four sampling Plots, arranged parallel to the waterline and spaced at 10 m. The temporal and spatial factors were arranged orthogonally (Fig. 2). Season and Condition were fixed, and all the other factors were random.

Sampling was performed in 2011 during austral summer (January and February) and winter (June and July), always at low spring tides. From each plot ($1^2$ m), three sediment samples were collected for faunal abundance and richness analysis, and one sample was collected for environmental variables analysis (redox potential discontinuity layer, grain-size distribution (according to

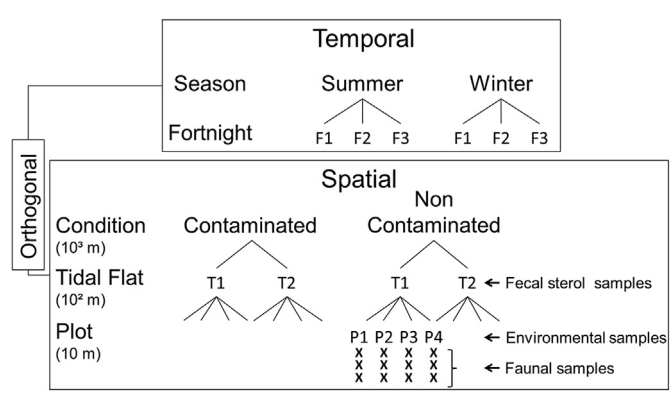


**Fig. 2.** Sampling design diagram. Temporal and spatial scales correspond to the factors of the linear model. Two Seasons were included (Summer and Winter), with three consecutive Fortnights per Season (F1, F2 and F3). In each Fortnight, two Conditions were sampled (Contaminated and Non-Contaminated) $10^3$; m apart, with two Tidal flats per Condition (T1 and T2) $10^2$ m apart, four Plots per Tidal flat (P1, P2, P3 e P4) (10 m apart) and three replicates each ($1^2$ m plot).

Suguio, 1973), calcium carbonate, organic matter, total organic carbon, total nitrogen and total phosphorus). Environmental samples were collected by scooping the surficial layer of sediment, conserved in plastic containers, labeled and kept on ice until transferred to the laboratory.

Faunal samples were collected using a PVC corer (diameter = 10 cm, depth = 10 cm). Samples were fixed with 6% formaldehyde buffered with sodium tetraborate, washed through a 0.5 mm sieve and preserved in 70% alcohol with Rose Bengal dye. Organisms were sorted and identified to the lowest taxonomic resolution.

Grain-size analyses were performed using the hand-sieving method for sands (grain-size > 63 μm) and the pipette sampling method for sediment fractions finer than 63 μm. Calcium carbonate ($CaCO_3$) and organic matter concentrations were determined using the gravimetric method after acid digestion with 1 mol $L^{-1}$ hydrochloric acid and combustion in a furnace at 550 °C for 1 h, respectively. Total nitrogen (TN) and total phosphorus (TP) concentrations were determined according to Grasshoff et al. (1999), and total organic carbon (TOC) concentrations were determined according to Strickland and Parsons (1972).

A single sample was taken from each tidal flat at each sampling event (fortnight) for the fecal sterol analysis, following routines described by Kawakami and Montone (2002). Sediments were extracted in a Soxhlet apparatus, and the steroid 5α-cholestane was added as a surrogate. The extract was concentrated using rotoevaporation and cleaned up using column chromatography with deactivated alumina and ethanol. The extracts were then evaporated to dryness and derivatized with BSTFA (bis (trimethylsilyl) trifluoroacetamide) with 1% TMCS (trimethylchlorosilane).

The organic extracts were analyzed using gas chromatography with an Agilent HP 7890A coupled with a flame ionization detector (FID). Instrument specifications and calibration procedures are described by (Montone et al., 2010). Procedural blanks were performed for each series of 10 extractions, and the interfering peaks did not interfere with the analyses of the target compounds. Internal standard recoveries ranged from 72 to 103% (mean 89.2 ± 11.7%). The detection limits (DLs) were <0.01 μg $g^{-1}$ for all analyzed compounds. Measured concentrations of target steroids in the IAEA-417 reference material were within 90–110% of the certified values provided by the International Atomic Energy Agency (IAEA).

### 2.3. Data analysis

Statistical analyses were done using the software PRIMER 6 (Clarke and Gorley, 2006). with the PERMANOVA + add-on (Anderson et al., 2008) and the software R(R Core Team, 2013). Spatial and temporal variability in benthic macrofauna was evaluated at multiple scales using the following linear mixed model:

$$X = \mu + S_i + F(S)_{j(i)} + C_k + T(C)_{l(k)} + P(T(C))_{m(l(k))} + SC_{ik} + ST(C)_{il(k)} + SP(T(C))_{im(l(k))} + F(S)C_{j(i)k} + F(S)T(C)_{j(i)l(k)} + F(S)P(T(C))_{j(i)m(l(k))} + e_{n(ijklm)}$$

where S = Seasons; F(S) = Fortnights, nested in Seasons; C = Conditions; T(C) = Tidal flats, nested in Conditions; P(T(C)) = Plots, nested in Tidal flats.

Due to the lack of replication at the Plot scale ($10^1$ m), the spatial scales tested on the linear model for the environmental variables were only Condition ($10^3$; m) and Tidal Flats ($10^2$ m), as follows:

$$X = \mu + S_i + F(S)_{j(i)} + C_k + T(C)_{l(k)} + SC_{ik} + ST(C)_{il(k)} + F(S)C_{j(i)k} + F(S)T(C)_{j(i)l(k)} + e_{n(ijkl)}$$

where S = Seasons; F(S) = Fortnights, nested in Seasons; C = Conditions; T(C) = Tidal flats, nested in Conditions.

Canonical Analysis of Principal Coordinates (CAP), a constrained ordination procedure (Anderson and Willis, 2003; Anderson, 2004), was performed in order to identify relationships between chemical (TN, OM, TP, RPD, TOC, Mud) and abundance of the dominant taxa, which achieved more than 90% of the total abundance, based on Bray–Curtis dissimilarity matrix. The hypotheses were tested by obtaining a p-value with permutation procedures (9999 permutations) on the canonical test statistics (squared canonical correlations). The relative contribution of the dominant taxa to the differences found was assessed by the correlation coefficient resulting between each species and the canonical axis.

Univariate analysis separately examined total abundance (N) and number of species (S) and the relative abundance of the six dominant taxa, which comprised 80% of the total abundance (Table 3). The linear model was tested using a permutational analysis of variance, sometimes referred to as PERANOVA (e.g., Ezgeta-Balić et al., 2011; Fanelli et al., 2011; Sweeting et al., 2009) based on the Euclidean distance measure (Anderson, 2001). The use of Euclidean distance as a measure of association makes this univariate test similar to a traditional ANOVA (Anderson et al., 2008). When necessary, the variables were transformed to fulfill linear model assumptions.

The multivariate analysis included an nMDS (Non-metric Multidimensional Scaling) ordination (based on a Bray–Curtis dissimilarity matrix) and permutational multivariate analyses of variance (PERMANOVAs). The latter analyses were used to test hypotheses concerning spatial and temporal patterns observed in the structure of infaunal assemblages as a whole.

Of the 11 terms generated by the linear model, five could not be directly tested due to the complexity of the design, which makes it impossible to identify the appropriate denominator mean square for F ratios calculation. For these terms, namely PE, CO, QU(PE), BA(CO), PExCO, PO(BA(CO)) and PExBA(CO), a linear combination of mean squares were constructed using the procedure described by Satterthwaite (1946), detailed by Blackwell et al. (1991) and implemented in PERMANOVA+. The linear combination of effects may induce to an unknown F-distribution under a true null hypothesis. However, the permutation method avoids that problem so the p-value can be used for valid inference (Anderson et al., 2008). The high number of variation sources inherent to our linear model could raise the occurrence of type I error. Estimates of components of variation were also calculated to evaluate the amount of variation attributed to each source, specially to the residuals, and were analyzed together with the analysis of variance. Whenever the estimate of the component of variation for some term in the model was negative, its value was set to zero since the term in question had a large p-value.

## 3. Results

Coprostanol concentrations varied between < DL and 14 μg $g^{-1}$ (Fig. 3). The values of the ratio coprostanol/coprostanol + cholestanol were below 0.3 for all samples at the Non-Contaminated site. In contrast, values ranged between 0.48 and 0.85 at Contaminated sites, providing a preliminary diagnostic of sewage contamination. The values of the ratio coprostanol/coprostanol + cholesterol for the Non-Contaminated site were less than 0.15, remaining below the threshold value established for contaminated areas. At the Contaminated site, the values varied between 0.12 and 0.77, and only the samples from Tidal Flat 1 showed values close to or higher than 0.5, an indication of sewage input. The ratio of epicoprostanol to coprostanol was <0.2 at the Contaminated site, indicating the presence of raw sewage

**Table 1**
Summary of PERANOVA: degrees of freedom (df), pseudo-F (Ps-F), p-values (p) and components of variation (CV-%) for total nitrogen (TN), total phosphorus (TP), total organic carbon (TOC), organic matter (OM), redox potential discontinuity layer (RPD) and percentage of silt and clay (Mud). S = Seasons, F(S) = Fortnights, C = Conditions, and T(C) = Tidal flats.

| | | TN | | | TP | | | TOC | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Source | *df* | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV |
| S | 1 | 1.02 | 0.51 | 1.5 | 0.93 | 0.56 | 0 | 1.21 | 0.42 | 3.2 |
| F(S) | 4 | 2.21 | 0.16 | 10.5 | 1.40 | 0.32 | 5.4 | 0.78 | 0.57 | 0 |
| C | 1 | 0.79 | 0.64 | 0 | 2.07 | 0.19 | 16.4 | 0.18 | 0.99 | 0 |
| T(C) | 2 | 3.13 | 0.10 | 11.4 | 8.46 | **0.01** | 19.1 | 5.81 | **0.03** | 23.1 |
| SxC | 1 | 0.86 | 0.60 | 0 | 0.53 | 0.81 | 0 | 16.49 | **<0.01** | 20.4 |
| SxT(C) | 2 | 1.83 | 0.22 | 10.1 | 3.67 | 0.07 | 16.1 | 0.13 | 0.88 | 0 |
| F(S)xC | 4 | 5.87 | **0.02** | 29.8 | 1.87 | 0.21 | 11.3 | 0.11 | 0.97 | 0 |
| F(S)xT(C) | 8 | 3.43 | **<0.01** | 16.1 | 4.17 | **<0.01** | 14.9 | 2.31 | **0.03** | 19.4 |
| Residual | 72 | | | 20.6 | | | 16.8 | | | 33.9 |
| | | OM | | | RPD | | | Mud | | |
| Source | *df* | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV |
| S | 1 | 0.97 | 0.54 | 0 | 0.33 | 0.93 | 0 | 2.87 | 0.11 | 11.3 |
| F(S) | 4 | 4.96 | **0.03** | 22.4 | 2.86 | 0.10 | 13.1 | 2.27 | 0.15 | 7.1 |
| C | 1 | 0.35 | 0.92 | 0 | 12.60 | **<0.01** | 32.8 | 0.08 | 1.00 | 0 |
| T(C) | 2 | 2.82 | 0.12 | 12.4 | 1.57 | 0.27 | 5.9 | 27.12 | **<0.01** | 26.2 |
| SxC | 1 | 1.05 | 0.49 | 3.2 | 0.59 | 0.77 | 0 | 0.66 | 0.73 | 0 |
| SxT(C) | 2 | 1.83 | 0.22 | 11.8 | 0.72 | 0.52 | 0 | 2.92 | 0.11 | 10.1 |
| F(S)xC | 4 | 0.51 | 0.73 | 0 | 1.43 | 0.31 | 8.9 | 3.33 | 0.07 | 13.6 |
| F(S)xT(C) | 8 | 1.39 | 0.21 | 12.0 | 2.50 | **0.02** | 14.9 | 0.62 | 0.76 | 0 |
| Residual | 72 | | | 38.2 | | | 24.4 | | | 31.8 |

p values <0.05 are shown in bold.

**Table 2**
Summary of PERMANOVA and components of variation (CV-%) for the macrobenthic assemblages. S = Seasons, F(S) = Fortnights, C = Conditions, T(C) = Tidal flats and P (T(C)) = Plots.

| Source | *df* | F | *p* | CV |
|---|---|---|---|---|
| S | 1 | 2.97 | **0.01** | 13.2 |
| F(S) | 4 | 6.40 | **<0.01** | 12.1 |
| C | 1 | 2.92 | **0.02** | 11.8 |
| T(C) | 2 | 4.55 | **<0.01** | 8.5 |
| P (T(C)) | 12 | 1.45 | **<0.01** | 1.1 |
| SxC | 1 | 1.61 | 0.12 | 3.6 |
| SxT(C) | 2 | 1.88 | **0.01** | 4.2 |
| SxP (T(C)) | 12 | 1.40 | **0.01** | 2.0 |
| F(S)xC | 4 | 1.34 | 0.12 | 1.5 |
| F(S)xT(C) | 8 | 2.42 | **<0.01** | 5.3 |
| F(S)xP (T(C)) | 48 | 1.34 | **<0.01** | 3.8 |
| Residual | 192 | | | 33.0 |

p values <0.05 are shown in bold

contamination of the sediment (Fig. 3). Sediment texture was homogeneous at the scales investigated, with the prevalence of poorly sorted to well-sorted fine sand, and the results are not displayed here for shortness.

At the Contaminated site, total nitrogen (TN) levels were higher, especially during Summer, with significant interactions at the smallest scales (Fortnight × Condition and Fortnight × Tidal Flat) (Fig. 4, Table 1). The levels increased along Fortnights at Tidal Flat 1 and decreased at Tidal Flat 2. The Fortnight × Tidal Flat interaction was also significant for total phosphorus (TP), total organic carbon (TOC) and redox potential discontinuity layer (RPD) depth (Table 1).

At the Non-Contaminated site, TP concentrations were consistent across Tidal Flat 2, whereas at Tidal Flat 1 they decreased from Summer to Winter. TOC levels varied differently within Conditions among Seasons with smaller concentrations on summer than winter (Fig. 4). The RPD was smaller at the Contaminated site, remaining almost all time below 1 cm. Organic matter levels varied significantly between Fortnights in both Conditions. Fine sediment content was homogeneous across the Contaminated site and varied among Tidal Flats at Non-Contaminated Site (Table 1 and Fig. 4). The fortnight x tidal flat interaction was the main significant source of variation for the environmental variables (indicated by p-values). Nevertheless, the components of variation showed that the most important scale of variability for TN was fortnight x condition, while TP varied consistently among tidal flats and RPD between conditions. For TOC, OM and mud, the components of variation were larger at the residual scale, indicating high variability within replicates (Table 1).

The faunal assemblage was dominated by oligochaetes (62%), polychaetes (38%) and molluscs (17%). nMDS ordinations separated the fauna into groups corresponding to summer and winter and the different conditions within each season (Fig. 5). PERMANOVA showed significant differences for the main factors (Season and Condition) and spatio-temporal interactions for most factors, except for Season x Condition and Fortnight x Condition, meaning that faunal differences between conditions were not relevant at the analyzed temporal scales (Table 2). Regardless the significance of the lower-order interactions, these factors did not have high components of variation. As observed in the grouping pattern of the nMDS, the greatest variation occurred among Conditions (11.8%), Seasons (13.2%) and Fortnights (12.1%) .

CAP analysis revealed a separation between Conditions at axis 2 ($\delta = 0.62$) with TP, TN, *P. frici*, *Laeonereis culveri* and *Capitella* sp. correlated to Contaminated sites while RPD, H. australis and *Sigambra* sp. correlated with Non-Contaminated sites. There was a relationship between changes in assemblage structure and MUD, OM and TOC among Seasons at axis 1 with a canonical correlation (using $m = 6$ principal coordinate axes) of $\delta = 0.75$ (Fig. 6). OM, MUD and TP presented higher levels in winter, while TN contents were higher in summer.

Significant differences in abundance were observed among Fortnights and Tidal Flats. The interactions between Season x Condition, Fortnight x Tidal Flat and Fortnight x Plot were also significant (Table 3). The differences in total abundance between Tidal Flats were especially pronounced at the Contaminated site, with higher abundances in Tidal Flat 1. Furthermore, the Fortnight 3

**Table 3**
Summary of PERANOVA: degrees of freedom (df), pseudo-F (Ps-F), p-values (p) and components of variation (CV-%) for total abundance, number of species and numerically dominant species for the spatial, temporal and interactive scales. S = Seasons, F(S) = Fortnights, C = Conditions, T(C) = Tidal flats and P (T(C)) = Plots.

| | | Abundance | | | Number of species | | | Tubificinae sp | | | *L. culveri* | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Source | *Df* | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV |
| S | 1 | 1.84 | 0.24 | 8.3 | 14.90 | **<0.01** | 30.2 | 0.50 | 0.84 | 0 | 1.41 | 0.35 | 4.5 |
| F(S) | 4 | 5.88 | **0.02** | 14.0 | 20.70 | **<0.01** | 13.2 | 4.81 | **0.03** | 14.8 | 5.04 | **0.03** | 6.0 |
| C | 1 | 1.90 | 0.22 | 13.5 | 2.09 | 0.19 | 4.7 | 3.27 | 0.08 | 22.2 | 2.15 | 0.18 | 16.6 |
| T(C) | 2 | 11.30 | **<0.01** | 18.8 | 1.72 | 0.09 | 3.2 | 8.82 | **<0.01** | 18.2 | 45.70 | **<0.01** | 21.4 |
| P (T(C)) | 12 | 1.83 | 0.07 | 3.7 | 2.28 | **0.02** | 4.4 | 0.73 | 0.71 | 0 | 1.67 | 0.10 | 2.7 |
| SxC | 1 | 4.10 | **0.05** | 9.5 | 4.79 | **0.04** | 10.2 | 0.83 | 0.61 | 0 | 1.69 | 0.27 | 7.4 |
| SxT(C) | 2 | 0.38 | 1.00 | 0 | 1.06 | 0.46 | 1.2 | 0.22 | 1.00 | 0 | 5.50 | **<0.01** | 10.7 |
| SxP (T(C)) | 12 | 1.00 | 0.44 | 0.9 | 1.56 | 0.14 | 4.1 | 0.54 | 0.88 | 0 | 2.66 | **0.01** | 5.9 |
| F(S)xC | 4 | 0.80 | 0.56 | 0 | 3.20 | 0.08 | 6.3 | 1.70 | 0.24 | 9.0 | 2.16 | 0.16 | 4.5 |
| F(S)xT(C) | 8 | 6.48 | **<0.01** | 11.6 | 1.60 | 0.15 | 3.6 | 7.85 | **<0.01** | 14.2 | 2.23 | **0.04** | 4.4 |
| F(S)xP (T(C)) | 48 | 1.61 | **0.01** | 6.1 | 1.25 | 0.15 | 4.2 | 1.54 | **0.02** | 6.4 | 1.22 | 0.18 | 3.4 |
| Residual | 192 | | | 13.6 | | | 14.6 | | | 15.2 | | | 12.5 |

| | | *H. australis* | | | *Paranais* cf. *frici* | | | *Sigambra* sp. | | | *Capitella* sp. | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Source | *Df* | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV | *Ps*-F | *p* | CV |
| S | 1 | 0.14 | 1.00 | 0 | 1.72 | 0.26 | 10.9 | 3.80 | 0.06 | 15.3 | 5.70 | **0.02** | 6.8 |
| F(S) | 4 | 10.30 | **<0.01** | 18.5 | 0.86 | 0.53 | 0 | 27.80 | **<0.01** | 13.4 | 0.30 | 0.88 | 0 |
| C | 1 | 6.47 | **0.02** | 11.6 | 1.16 | 0.45 | 5.2 | 1.30 | 0.40 | 4.4 | 2.70 | 0.12 | 17.2 |
| T(C) | 2 | 0.54 | 0.96 | 0 | 5.68 | **<0.01** | 15.6 | 6.10 | **<0.01** | 10.2 | 4.00 | **<0.01** | 16.0 |
| P (T(C)) | 12 | 2.72 | **0.01** | 7.9 | 2.23 | **0.03** | 3.0 | 1.60 | 0.13 | 4.8 | 5.10 | **<0.01** | 7.4 |
| SxC | 1 | 7.62 | **0.01** | 15.8 | 1.16 | 0.45 | 7.3 | 0.10 | 1.00 | 0 | 6.10 | **0.02** | 9.7 |
| SxT(C) | 2 | 0.43 | 0.99 | 0 | 5.62 | **<0.01** | 22.0 | 3.30 | **<0.01** | 8.6 | 0.10 | 1.00 | 0 |
| SxP (T(C)) | 12 | 1.08 | 0.39 | 2.5 | 2.32 | **0.02** | 4.4 | 1.20 | 0.32 | 3.9 | 2.10 | **0.03** | 5.5 |
| F(S)xC | 4 | 1.29 | 0.35 | 4.7 | 0.98 | 0.47 | 0 | 7.90 | **0.01** | 9.6 | 0.30 | 0.89 | 0 |
| F(S)xT(C) | 8 | 2.69 | **0.02** | 9.6 | 25.80 | **<0.01** | 16.6 | 0.40 | 0.89 | 0 | 20.50 | **<0.01** | 19.8 |
| F(S)xP (T(C)) | 48 | 1.67 | **0.01** | 9.4 | 0.60 | 0.98 | 0 | 2.10 | **<0.01** | 11.2 | 0.80 | 0.83 | 0 |
| Residual | 192 | | | 19.9 | | | 14.9 | | | 18.6 | | | 17.5 |

p values <0.05 are shown in bold

had the lowest abundance, regardless of Condition or Season (Fig. 7).

Temporal scales were an important source of variation for the number of species. Significant effects were found among Seasons, Fortnights and Plots, and for the interactions Season x Condition and Fortnight x Condition (Table 3). Graphically, the number of species increased from summer to winter and, as seen in the total abundance, Fortnight 3 had the lowest number of species (Fig. 7).

The most abundant taxa were Tubificinae sp. (39.3%), *L. culveri* (13.8%), *Heleobia australis* (10.6%), *Paranais* cf. *frici* (9.7%), *Sigambra* sp. (4.8%) and *Capitella* sp. (2.8%). Tubificinae sp., *L. culveri*, and *Capitella* sp. had a similar pattern, occurring mainly at the Contaminated site with higher abundances at Tidal Flat 1. The differences between tidal flats decreased along seasons and were smallest on Fortnight 3 (Fig 7).

During the summer, *H. australis* was found mainly at the Non-Contaminated site, with higher abundances at Tidal Flat 1. In winter, *H. australis* occurred in both conditions, especially on Fortnights 1 and 2, with a decrease on Fortnight 3 (Fig. 7). The differences among Fortnights, Conditions and Plots and the Season x Condition and Fortnight x Tidal Flat interactions were significant and the Fortnight scale had the larger component of variation (Table 3). *Paranais* cf. *frici* occurred almost exclusively at the Contaminated Tidal Flat 1 during winter (Fig. 7). Consequently, significant differences were found at the tidal flat scale and almost all interactions between tidal flat and the temporal scales. *Sigambra* sp. was found in all samples, with abundance slightly lower at the Contaminated site and during summer (Fig. 7). There were differences between Tidal Flats for all sampling events, except for the Contaminated site during Summer and Fortnight 3 which had the lowest abundances for all sampling events.

In general, factors Tidal Flat, Fortnight and Fortnight x Tidal Flat were frequently associated to significant differences, meaning that most differences in faunal assemblages were found at the scale of hundreds of meters, at the within-site scale and the smallest temporal scale, i.e., the Fortnight scale.

## 4. Discussion

Coprostanol is used as an indicator of sewage sources in coastal areas, with concentrations higher than 0.5 μg $g^{-1}$ being indicative of contamination (González-Oreja and Saiz-Salinas 1998). The levels of coprostanol found at Paranaguá Bay were higher than those found in some much more densely populated estuaries, such as Santos Bay, Brazil, with maximum values of 8.5 μg $g^{-1}$ (Martins et al., 2008) and lower than those found at Guanabara Bay, Brazil, with a maximum recorded value of 40 μg $g^{-1}$ (Carreira et al., 2004); Río de la Plata, Uruguay (maximum of 21.2 μg $g^{-1}$) (Venturini et al., 2015) and Mondego River Estuary, Portugal (maximum of 16.6 μg $g^{-1}$) (Rada et al., 2016).

Previous regional studies found maximum values of 2.2 μg $g^{-1}$ in sediments near the Paranaguá port, 0.65 μg $g^{-1}$ in the subtidal zone close to the sampling site of this study and 1.69 μg $g^{-1}$ in the Cotinga sub-estuary (Abreu-Mota et al., 2014; Martins et al., 2010). Although it may be unexpected given the relatively small population of Paranaguá City (140,000 inhabitants) and the previously reported low coprostanol levels, the high values recorded in this study occurred in three consecutive samplings, representing a consistent result and excluding the possibility of sampling and laboratory artifacts. Such high values may reflect low rainfall during the sampling period, which could have decreased the dispersion and dilution rates of the effluents discharged into the Itiberê River (Martins et al., 2010).

Most of the variability related to the magnitude of effects was concentrated at the larger spatial scales, Conditions ($10^3$; meters) and Tidal Flats ($10^2$ m) and the smaller temporal scales (Fortnights).

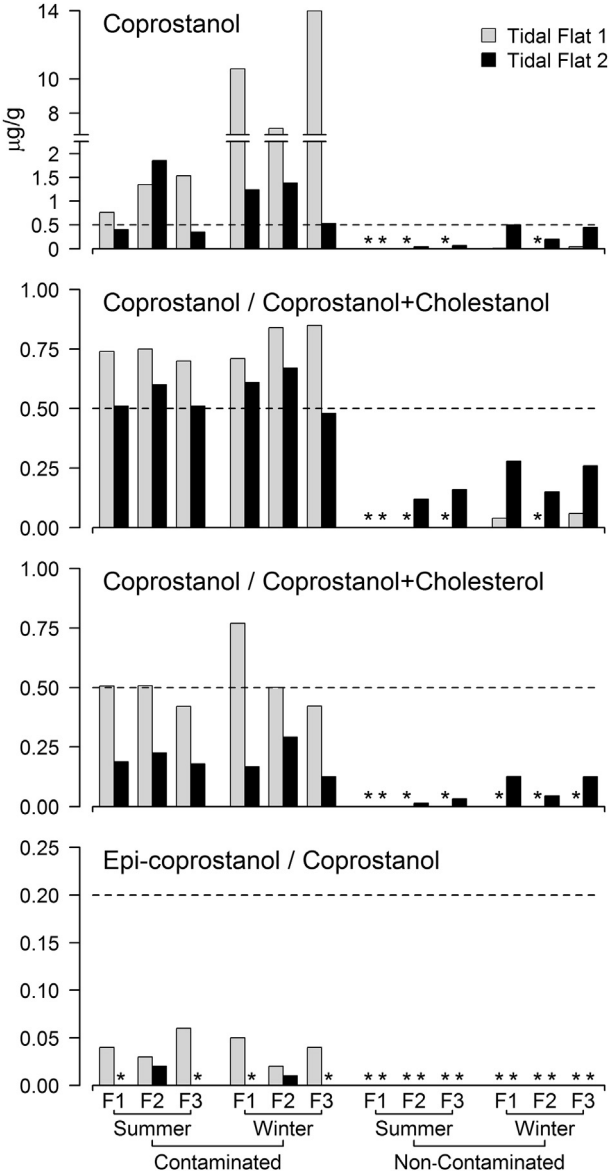


**Fig. 3.** Coprostanol levels (µg $g^{-1}$) and proportion of sterols in sediments sampled according to the following scales: Tidal Flats (gray and black bars), Fortnights (F1, F2 and F3), Seasons (Summer and Winter) and Conditions (Contaminated and Non-Contaminated). The dashed lines indicate the sewage contamination thresholds suggested by the literature (Grimalt et al., 1990; González-Oreja and Saiz-Salinas 1998, Mudge and Seguel, 1999; Peng et al., 2005). * = uncalculated, compounds below detection limits (<0.01 µg $g^{-1}$).

As previously reported by Souza et al. (2013), the local discharges of urban sewage were the major drivers acting at the scale of kilometers, evidenced by the high abundance of indicator species such as Tubificinae sp, *Paranais* cf. *frici*, *Capitella* sp., and *L. culveri*, which are recognized as tolerant to organic enrichment (Omena et al., 2012; Pires-Vanin et al. 2011). At the Contaminated site, the variability at the scale of Tidal Flats ($10^2$ m) was related to heterogeneity between tidal flats. In this Condition, Tidal flat 1 exhibited higher contamination levels and lower redox potential discontinuity layer in comparison to Tidal Flat 2, which may have caused the variations found on this scale. The high abundance of tubificids in all contaminated sites agree with their general opportunistic life strategy, including rapid growth, direct development and small size, which allow for their persistence in organically enriched habitats (Arimoro et al., 2007; Gamito, 2008).

The variability at the Tidal Flat ($10^2$ m) scale was also important at the Non-Contaminated site, possibly driven by variations in grain size and intrinsic characteristics of the local populations (Souza et al., 2013). Although there were no indications of contamination at the Non-Contaminated sites, contents of coprostanol, phosphorus, carbon, organic matter and fine sediments were higher in Tidal Flat 2, compared to Tidal Flat 1, which contributed to the heterogeneity between Tidal Flats at the Non-Contaminated site. However, these contents still do not indicate contamination itself at the Non-Contaminated sites, meaning that the differences between Conditions were more important than the variability within this scale.

Faunal differences between seasons were a possible consequence of desiccation due to high temperatures and solar exposure, or higher rainfall rates during summer. Mass mortality events in benthic populations have been locally reported (Pagliosa and Lana, 2000). These seasonal changes can cause high mortality and remove species less adapted to extreme conditions (Dauer, 1984). The larger incidence of tropical storms during summer may also destabilize the substrate, leading to resuspension and passive transport of surface-dwelling populations (Bouma et al., 2001; Grant et al., 1997; Pilditch et al., 2008). Moreover, such temporal variability is also attributed to the constant structural rearrangement of macrofaunal communities (Negrello-Filho and Lana, 2013).

Although apparently chaotic, and often regarded as simple uninteresting "noise", small-scale variability is ecologically relevant (Carroll, 2003). Large small-scale variability results from specific ecological processes operating to create microhabitats, possibly related to small-scale changes in environmental conditions, behavior and competition (Sandrini-Neto and Lana, 2012). In this way, high residual variance could indicate several factors and processes acting on scales not assessed at the present study.

The reduction of abundance and number of species on Fortnight 3 led to high variation at the smallest temporal scales. Sampling frequency and possible variations in the volume of samples (approximately 4% of the plot area) were not expected to have significant impacts on the macrofauna. The resilience of local tidal flat fauna to small-scale impacts is reportedly high (Bolam et al., 2004; Egres et al., 2012; Faraco and Lana, 2003; Gern and Lana, 2013). On the other hand, physical disturbance due to sampling may persist for at least two weeks and such changes may vary in unpredictable ways, not necessarily leading to mortality (Morrisey et al., 1996; Skilleter, 1996; Lindegarth and Underwood 2002).

Variations at the Fortnight scale were observed for total phosphorus, organic carbon and redox potential discontinuity layer, meaning that these variables could be acting as faunal structurers in addition to the natural rearrangement of communities. Fluctuations in organic carbon content may affect food availability while the redox potential discontinuity layer directy alterates the ph profile of the sediment as a possible response to low oxygen levels in subsurface layers (Rosenberg, 2001). Differences in the reproductive cycle, life strategy, and behavior play an important role in species responses to environmental heterogeneity at distinct spatio-temporal scales (Thrush et al., 1997). Considering logistic issues intrinsic to field surveys on intertidal flats, Fortnights was adopted as the finest temporal scale, which were not sufficient to attribute the environmental driving factors operating at fortnight scale. Marked short-scale variations in benthic populations within days or weeks are related to migration, predation or post-reproductive mortality (Fischer, 1999; Norkko et al., 2001). In this way, in order to identify the main driving factors at this scale we would need a smaller temporal scale to be nested within Fortnights, such as consecutive days sampled in each spring tide.

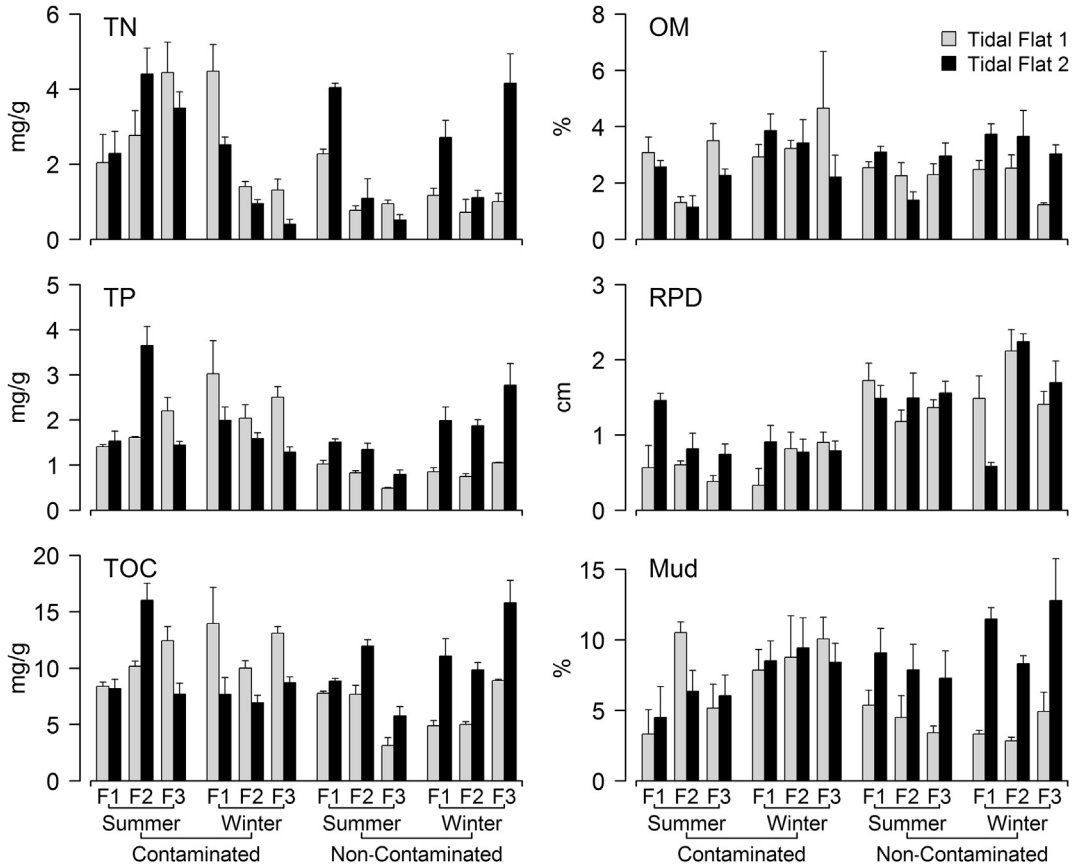


**Fig. 4.** Means (±S.E., n = 4) of total nitrogen (TN), total phosphorus (TP), total organic carbon (TOC), organic matter (OM), redox potential discontinuity layer (RPD) and percentage of silt and clay (Mud) in sediment sampled according to the following scales: Tidal Flats (gray and black bars), Fortnights (F1, F2 and F3), Seasons (Summer and Winter) and Conditions (Contaminated and Non-Contaminated).

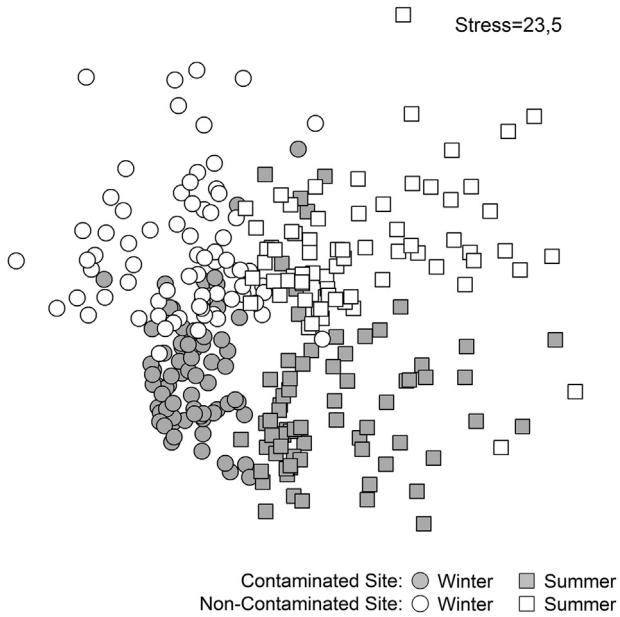


**Fig. 5.** Non-metric multidimensional scaling (nMDS) applied to the Bray–Curtis dissimilarity among species, showing the differences in the macrobenthic assemblages found at the Contaminated site (gray symbols) and the Non-Contaminated site (white symbols) in addition to between the summer (squares) and the winter (circles).

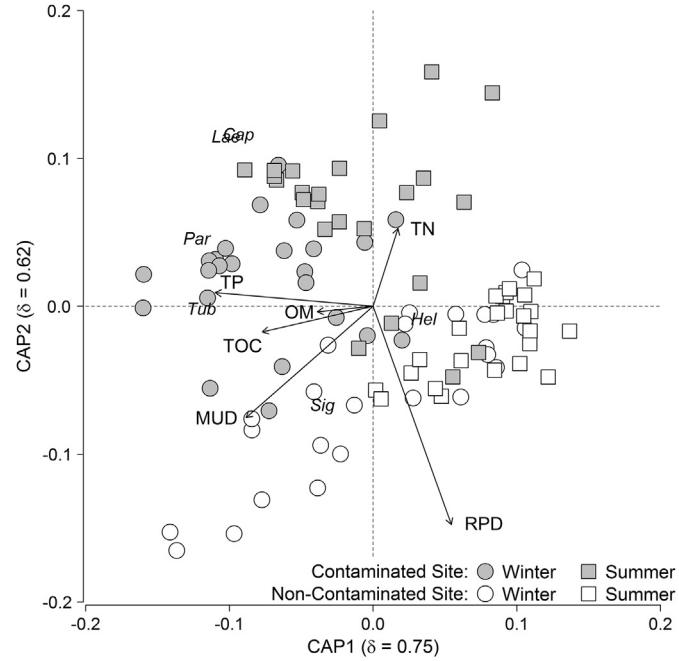


**Fig. 6.** Canonical Analysis of Principal Coordinates (CAP) between chemical (TN, OM, TP, RPD, TOC, and Mud) and most abundant taxa (Tubificinae sp., Laeonereis culveri, Heleobia australis, Paranais cf. frici, Sigambra sp. and Capitella sp.).

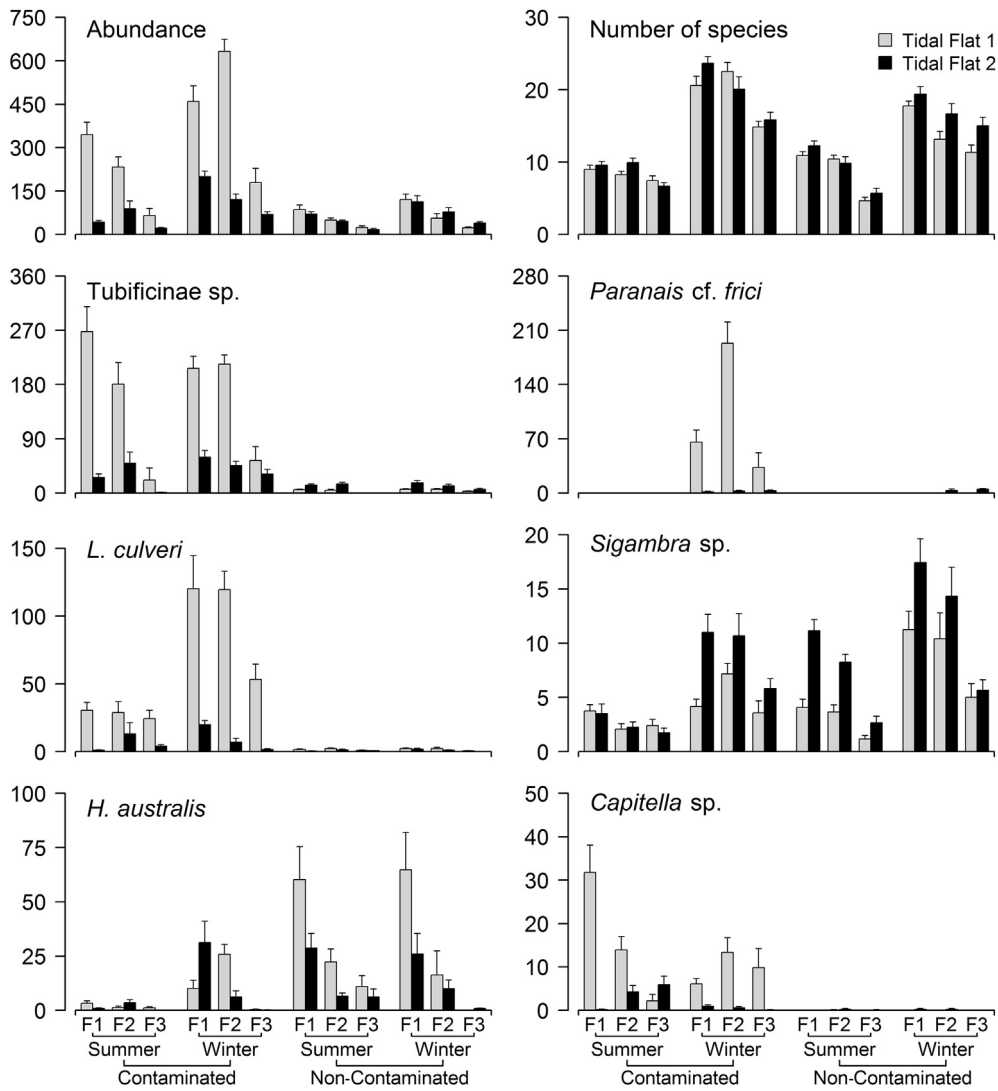


**Fig. 7.** Means (± S.E., n = 12) of macrofaunal total abundance, number of species, Tubificinae sp., Laeonereis culveri, Heleobia australis, Paranais cf. frici, Sigambra sp. and Capitella sp. associated to the spatial and temporal scales: Tidal flats (gray and black bars), Fortnights (F1, F2 and F3), Seasons (Summer and Winter) and Conditions (Contaminated and Non-Contaminated).

## 5. Conclusion

In general, temporal variability patterns were similar between Contaminated and Non-Contaminated sites and are possibly regulated by simultaneous processes operating under different conditions at scales greater than thousands of meters. Nevertheless, the large number of significant effects of interactions between Fortnight and Tidal Flat scales shows that short-term temporal variability may also reflect the degree of heterogeneity within individual sites.

Hierarchical experimental designs have previously addressed the importance of small-scale temporal variability in intertidal habitats (Murphy et al., 2009; Underwood, 1994). Our findings also highlight the need for much improved experimental designs to understand small-scale temporal variability of ecological communities. Despite the expected influence of sewage input on macrofaunal structure, variation patterns also reflected significant small-scale temporal changes not directly related to contamination. Not understanding such small-scale patterns may lead to erroneous interpretations of ecological datasets and mask true sources of variation.

The fortnightly samplings guaranteed the appropriate measurement of seasonal differences. The large variability found at the fortnight scale reinforced the importance of incorporating smaller temporal scales to sampling designs, otherwise seasonal variations could not be properly identified against background noise, or be recognized as spurious. Not understanding such small-scale patterns may lead to erroneous interpretations of ecological datasets and mask true sources of variation (Underwood and Chapman, 2013).Our results provide valuable evidence for the appropriate scales to be incorporated into novel monitoring and impact assessment protocols.

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